

COURSE CODE/TITLE:	CPE212 Automating Server Management	SECTION:	CPE31S2
DATE PERFORMED:	9/17/2026	DATE SUBMITTED:	9//2026
NAME:	Aicon H. Keliste	INSTRUCTOR:	Engr. Jimlord Quejado

ACTIVITY NO. 8.1 Install, Configure, and Manage Availability Monitoring tools

1. PROCEDURE OUTPUT

1. Create a playbook that installs Nagios in both Ubuntu and CentOS. Apply the concept of creating Roles.
2. Describe how you did step 1. (Provide screenshots and explanations in your report. Make your report detailed such that it will look like a manual.)
3. Show an output of the installed Nagios for both Ubuntu and CentOS.
4. Make sure to create a new repository in GitHub for this activity.

Site.yml

```

GNU nano 6.2                                roles/nagios/tasks/main.yml *
---
- name: Installing nagios in UBuntu
  block:
    - name: Update apt cache
      ansible.builtin.apt:
        update_cache: true

    - name: Install nagios and plugins (ubuntu)
      ansible.builtin.apt:
        name:
          - nagios4
          - nagios-plugins-basic
        state: present
      when: ansible_facts['os_family'] == "Debian"

- name: Install nagios in Centos
  block:
    - name: Install epel repository
      ansible.builtin.yum:
        name: epel-release
        state: present

    - name: Install nagios and plugins (centos)
      ansible.builtin.yum:
        name:
          - nagios
          - nagios-plugins-all
        state: present
      when: ansible_facts['os_family'] == "Redhat"

```

Failed

```
PLAY [web_servers, db_servers] *****

TASK [Gathering Facts] *****
ok: [192.168.56.103]
ok: [192.168.56.104]
[DEPRECATION WARNING]: Distribution centos 9 on host 192.168.56.105 should use /usr/libexec/platform-python, but is using
/usr/bin/python for backward compatibility with prior Ansible releases. A future Ansible release will default to using the
discovered platform python for this host. See https://docs.ansible.com/ansible/2.10/reference_appendices/interpreter_discovery.html
for more information. This feature will be removed in version 2.12. Deprecation warnings can be disabled by setting
deprecation_warnings=False in ansible.cfg.
ok: [192.168.56.105]

TASK [nagios : Update apt cache] *****
[WARNING]: Updating cache and auto-installing missing dependency: python3-apt
fatal: [192.168.56.105]: FAILED! => {"changed": false, "cmd": "apt-get update", "msg": "[Errno 2] No such file or directory: b'apt-get'", "rc": 2}
changed: [192.168.56.103]
fatal: [192.168.56.104]: FAILED! => {"changed": false, "msg": "Failed to update apt cache: E:Release file for http://security.ubuntu.com/ubuntu/di
sts/jammy-security/InRelease is not valid yet (invalid for another 6d 9h 26min 51s). Updates for this repository will not be applied., E:Release f
ile for http://ph.archive.ubuntu.com/ubuntu/dists/jammy-updates/InRelease is not valid yet (invalid for another 6d 10h 31min 52s). Updates for thi
s repository will not be applied., E:Release file for http://ph.archive.ubuntu.com/ubuntu/dists/jammy-backports/InRelease is not valid yet (invali
d for another 6d 8h 37min 50s). Updates for this repository will not be applied.")

TASK [nagios : Install nagios and plugins (ubuntu)] *****
ok: [192.168.56.103]

TASK [nagios : Install epel repository] *****
fatal: [192.168.56.103]: FAILED! => [{"ansible_facts": {"pkg_mgr": "apt"}, "changed": false, "msg": ["Could not detect which major revision of yum
is in use, which is required to determine module backend.", "You should manually specify use_backend to tell the module whether to use the yum (yu
m3) or dnf (yum4) backend"]}]]

PLAY RECAP *****
192.168.56.103      : ok=3    changed=1    unreachable=0    failed=1    skipped=0    rescued=0    ignored=0
192.168.56.104      : ok=1    changed=0    unreachable=0    failed=1    skipped=0    rescued=0    ignored=0
192.168.56.105      : ok=1    changed=0    unreachable=0    failed=1    skipped=0    rescued=0    ignored=0
```

Failed on the CentOS

```
keliste@Keliste-Workstation: ~/CPE212_KELISTE/HOA8.1

by setting deprecation_warnings=False in ansible.cfg.
ok: [192.168.56.105]

TASK [nagios : Update apt cache] *****
skipping: [192.168.56.105]
changed: [192.168.56.104]
changed: [192.168.56.103]

TASK [nagios : Install nagios and plugins (ubuntu)] *****
skipping: [192.168.56.105]
ok: [192.168.56.103]
changed: [192.168.56.104]

TASK [nagios : Install epel repository] *****
skipping: [192.168.56.103]
skipping: [192.168.56.104]
ok: [192.168.56.105]

TASK [nagios : Install nagios and plugins (centos)] *****
skipping: [192.168.56.103]
skipping: [192.168.56.104]
fatal: [192.168.56.105]: FAILED! => {"changed": false, "failures": [], "msg": "Depsolve Error occu
red: \n Problem: package nagios-plugins-all-2.5-2.el9.x86_64 from epel requires nagios-plugins-dis
k_smb, but none of the providers can be installed\n - package nagios-plugins-disk_smb-2.5-2.el9.x
86_64 from epel requires perl(utf8::all), but none of the providers can be installed\n - conflict
ing requests\n - nothing provides perl(Import::Into) needed by perl-utf8-all-0.024-14.el9.noarch
from epel\n - nothing provides perl(PerlIO::utf8_strict) needed by perl-utf8-all-0.024-14.el9.noa
rch from epel", "rc": 1, "results": []}

PLAY RECAP *****
192.168.56.103      : ok=3    changed=1    unreachable=0    failed=0    skipped=2    rescue
d=0    ignored=0
192.168.56.104      : ok=3    changed=2    unreachable=0    failed=0    skipped=2    rescue
d=0    ignored=0
192.168.56.105      : ok=2    changed=0    unreachable=0    failed=1    skipped=2    rescue
d=0    ignored=0

keliste@Keliste-Workstation:~/CPE212_KELISTE/HOA8.1$
```

Add to site.yml

```
- hosts: web_servers, db_servers
  become: true
  roles:
    - nagios
```

[Wrote 121 lines]

```
^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^_ Go To Line
```

Failed on Centos

```
keliste@Keliste-Workstation: ... x keliste@Keliste-Workstation: ... x keliste@Keliste-Workstation: ~... x v
ok: [192.168.56.105]

TASK [nagios : Update apt cache] *****
skipping: [192.168.56.105]
changed: [192.168.56.104]
changed: [192.168.56.103]

TASK [nagios : Install nagios and plugins (ubuntu)] *****
skipping: [192.168.56.105]
ok: [192.168.56.104]
ok: [192.168.56.103]

TASK [nagios : Install epel repository] *****
skipping: [192.168.56.103]
skipping: [192.168.56.104]
ok: [192.168.56.105]

TASK [nagios : Install nagios and plugins (centos)] *****
skipping: [192.168.56.103]
skipping: [192.168.56.104]
fatal: [192.168.56.105]: FAILED! => {"changed": false, "failures": [], "msg": "Depsolve Error occurred: \n Problem: package nagios-plugins-all-2.5-2.el9.x86_64 from epel requires nagios-plugins-disk_smb, but none of the providers can be installed\n - package nagios-plugins-disk_smb-2.5-2.el9.x86_64 from epel requires perl(utf8::all), but none of the providers can be installed\n - conflicting requests\n - nothing provides perl(Import::Into) needed by perl-utf8-all-0.024-14.el9.noarch from epel\n - nothing provides perl(PerlIO::utf8_strict) needed by perl-utf8-all-0.024-14.el9.noarch from epel", "rc": 1, "results": []}

PLAY RECAP *****
192.168.56.102      : ok=8    changed=0    unreachable=0    failed=0    skipped=3    rescue
d=0    ignored=0
192.168.56.103      : ok=8    changed=1    unreachable=0    failed=0    skipped=5    rescue
d=0    ignored=0
192.168.56.104      : ok=8    changed=1    unreachable=0    failed=0    skipped=5    rescue
d=0    ignored=0
192.168.56.105      : ok=12   changed=1    unreachable=0    failed=1    skipped=5    rescue
d=0    ignored=0
```

Add on the site.yml

```

update_cache: true

- name: Install nagios and plugins (ubuntu)
  ansible.builtin.apt:
    name:
      - nagios4
      - nagios-plugins-basic
    state: present
  when: ansible_facts['os_family'] == "Debian"

- name: Install nagios in Centos
  block:
    - name: Install epel repository
      ansible.builtin.yum:
        name: epel-release
        state: present

    - name: Install nagios and plugins (centos)
      ansible.builtin.yum:
        name:
          - nagios
          - nagios-plugins-load
          - nagios-plugins-disk
          - nagios-plugins-users
          - nagios-plugins-procs
          - nagios-plugins-swap
          - nagios-plugins-ping
        state: present
  when: ansible_facts['os_family'] == "RedHat"
  
```

^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location
 ^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^_ Go To Line

Fixed

```

TASK [nagios : Update apt cache] *****
skipping: [192.168.56.105]
changed: [192.168.56.104]
changed: [192.168.56.103]

TASK [nagios : Install nagios and plugins (ubuntu)] *****
skipping: [192.168.56.105]
ok: [192.168.56.103]
ok: [192.168.56.104]

TASK [nagios : Install epel repository] *****
skipping: [192.168.56.103]
skipping: [192.168.56.104]
ok: [192.168.56.105]

TASK [nagios : Install nagios and plugins (centos)] *****
skipping: [192.168.56.103]
skipping: [192.168.56.104]
changed: [192.168.56.105]

PLAY RECAP *****
192.168.56.102      : ok=8    changed=0    unreachable=0    failed=0    skipped=3    rescue
d=0    ignored=0
192.168.56.103      : ok=8    changed=1    unreachable=0    failed=0    skipped=5    rescue
d=0    ignored=0
192.168.56.104      : ok=8    changed=1    unreachable=0    failed=0    skipped=5    rescue
d=0    ignored=0
192.168.56.105      : ok=13   changed=2    unreachable=0    failed=0    skipped=5    rescue
d=0    ignored=0
192.168.56.106      : ok=0    changed=0    unreachable=1    failed=0    skipped=0    rescue
d=0    ignored=0

keliste@Keliste-Workstation:~/CPE212_KELISTE/H0A8.1$
  
```

Verify Nagios in Ubuntu Server 1

```
keliste@Keliste-Workstation:~/CPE212_KELISTE$ ssh keliste@192.168.56.103
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-138-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

3 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

New release '24.04.5 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Last login: Sat Sep 19 10:08:23 2026 from 192.168.56.102
keliste@Keliste-Server1:~$ systemctl status nagios4
● nagios4.service - nagios4
   Loaded: loaded (/lib/systemd/system/nagios4.service; disabled; vendor preset: enabled)
   Active: active (running) since Thu 2026-09-17 09:56:31 +08; 2 days ago
     Docs: man:nagios4
   Process: 23344 ExecStartPre=sh -c nagiospipe=$(sed -n "s/^command_file=\\(.*\\)/\\1/p" ${NAGIOS}
  Main PID: 23346 (nagios4)
    Tasks: 6 (limit: 4598)
   Memory: 2.8M
      CPU: 7.946s
   CGroup: /system.slice/nagios4.service
           └─23346 /usr/sbin/nagios4 /etc/nagios4/nagios.cfg
             └─23347 /usr/sbin/nagios4 --worker /var/lib/nagios4/rw/nagios.qh
               └─23348 /usr/sbin/nagios4 --worker /var/lib/nagios4/rw/nagios.qh
                 └─23349 /usr/sbin/nagios4 --worker /var/lib/nagios4/rw/nagios.qh
                   └─23350 /usr/sbin/nagios4 --worker /var/lib/nagios4/rw/nagios.qh
                     └─23351 /usr/sbin/nagios4 --worker /var/lib/nagios4/rw/nagios.qh
                       └─23421 /usr/sbin/nagios4 /etc/nagios4/nagios.cfg

lines 1-16/16 (END)
```

Server 2

```
Connection to 192.168.56.103 closed.
keliste@Keliste-Workstation:~/CPE212_KELISTE$ ssh keliste@192.168.56.104
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-138-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

1 additional security update can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

New release '24.04.5 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Last login: Sat Sep 19 10:08:23 2026 from 192.168.56.102
keliste@Keliste-Server2:~$ systemctl status nagios4
● nagios4.service - nagios4
   Loaded: loaded (/lib/systemd/system/nagios4.service; disabled; vendor preset: enabled)
   Active: active (running) since Sat 2026-09-19 09:18:26 +08; 52min ago
     Docs: man:nagios4
   Process: 30474 ExecStartPre=sh -c nagiospipe=$(sed -n "s/^command_file=\\(.*\\)/\\1/p" ${NAGIOS4
 Main PID: 30476 (nagios4)
    Tasks: 6 (limit: 4598)
   Memory: 2.6M
      CPU: 2.122s
   CGroup: /system.slice/nagios4.service
           └─30476 /usr/sbin/nagios4 /etc/nagios4/nagios.cfg
           └─30477 /usr/sbin/nagios4 --worker /var/lib/nagios4/rw/nagios.qh
           └─30478 /usr/sbin/nagios4 --worker /var/lib/nagios4/rw/nagios.qh
           └─30479 /usr/sbin/nagios4 --worker /var/lib/nagios4/rw/nagios.qh
           └─30480 /usr/sbin/nagios4 --worker /var/lib/nagios4/rw/nagios.qh
           └─30510 /usr/sbin/nagios4 /etc/nagios4/nagios.cfg
```

Verify Nagios in Centos

```

Connection to 192.168.56.104 closed.
keliste@Keliste-Workstation:~/CPE212_KELISTE$ ssh keliste-centos@192.168.56.105
Last login: Thu Sep 17 11:52:04 2026 from 192.168.56.102
[keliste-centos@KelisteMain ~]$ systemctl status nagios
○ nagios.service - Nagios Core 4.4.14
   Loaded: loaded (/usr/lib/systemd/system/nagios.service; disabled; preset: disabled)
   Active: inactive (dead)
     Docs: https://www.nagios.org/documentation
[keliste-centos@KelisteMain ~]$ sudo systemctl enable --now nagios
[sudo] password for keliste-centos:
Sorry, try again.
[sudo] password for keliste-centos:
Created symlink /etc/systemd/system/multi-user.target.wants/nagios.service → /usr/lib/systemd/system/nagios.service.
[keliste-centos@KelisteMain ~]$ systemctl status nagios
● nagios.service - Nagios Core 4.4.14
   Loaded: loaded (/usr/lib/systemd/system/nagios.service; enabled; preset: disabled)
   Active: active (running) since Thu 2026-09-17 11:57:14 PST; 7s ago
     Docs: https://www.nagios.org/documentation
   Process: 47982 ExecStartPre=/usr/sbin/nagios -v /etc/nagios/nagios.cfg (code=exited, status=0>
   Process: 47984 ExecStart=/usr/sbin/nagios -d /etc/nagios/nagios.cfg (code=exited, status=0/SU>
  Main PID: 47986 (nagios)
    Tasks: 6 (limit: 10515)
   Memory: 6.5M (peak: 7.7M)
      CPU: 266ms
   CGroup: /system.slice/nagios.service
           └─47986 /usr/sbin/nagios -d /etc/nagios/nagios.cfg
             └─47987 /usr/sbin/nagios --worker /var/spool/nagios/cmd/nagios.qh
               └─47988 /usr/sbin/nagios --worker /var/spool/nagios/cmd/nagios.qh
                 └─47989 /usr/sbin/nagios --worker /var/spool/nagios/cmd/nagios.qh
                   └─47990 /usr/sbin/nagios --worker /var/spool/nagios/cmd/nagios.qh
                     └─47991 /usr/sbin/nagios -d /etc/nagios/nagios.cfg
lines 1-17/17 (END)

```

Github Repository

```

keliste@Keliste-Workstation:~/CPE212_KELISTE$ git add .
keliste@Keliste-Workstation:~/CPE212_KELISTE$ git status
On branch main
Your branch is up to date with 'origin/main'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    new file:   HOA8.1/files/CPE212_KELISTE
    new file:   HOA8.1/files/default_site.html
    deleted:    HOA8.1/roles/.roles.yml.swp
    modified:   HOA8.1/roles/nagios/tasks/main.yml
    new file:   HOA8.1/site.yml
    modified:   site.yml

keliste@Keliste-Workstation:~/CPE212_KELISTE$ git commit -m "FULL HOA8.1"
[main 0476260] FULL HOA8.1
 6 files changed, 241 insertions(+), 8 deletions(-)
 create mode 100644 HOA8.1/files/CPE212_KELISTE
 create mode 100644 HOA8.1/files/default_site.html
 delete mode 100644 HOA8.1/roles/.roles.yml.swp
 create mode 100644 HOA8.1/site.yml
keliste@Keliste-Workstation:~/CPE212_KELISTE$ git push -u origin main
Enumerating objects: 14, done.
Counting objects: 100% (14/14), done.
Delta compression using up to 2 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (8/8), 1.40 KiB | 477.00 KiB/s, done.
Total 8 (delta 2), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To github.com:AiconKeliste/CPE212_KELISTE.git
 6a48317..0476260  main -> main
Branch 'main' set up to track remote branch 'main' from 'origin'.
keliste@Keliste-Workstation:~/CPE212_KELISTE$

```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Answer the following:

1. What are the benefits of having an availability monitoring tool?

- An availability monitoring tool like Nagios checks whether servers and services are up and alerts admins the moment something goes down, instead of waiting for users to notice. This helps avoid downtime-related losses and lets issues get fixed before they cause bigger problems.

Conclusions/Learnings:

- Through this activity, I learned how to use Ansible roles to install and configure Nagios differently depending on the OS, using “when” conditions to separate the Ubuntu and CentOS tasks. I also ran into real errors along the way, like missing role directories, missing files, and dependency conflicts on CentOS, which taught me how to actually read and troubleshoot Ansible output instead of just running playbooks

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used:

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.



COMPUTER
ENGINEERING

QUEZON CITY

Purpose of Use:

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use:

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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